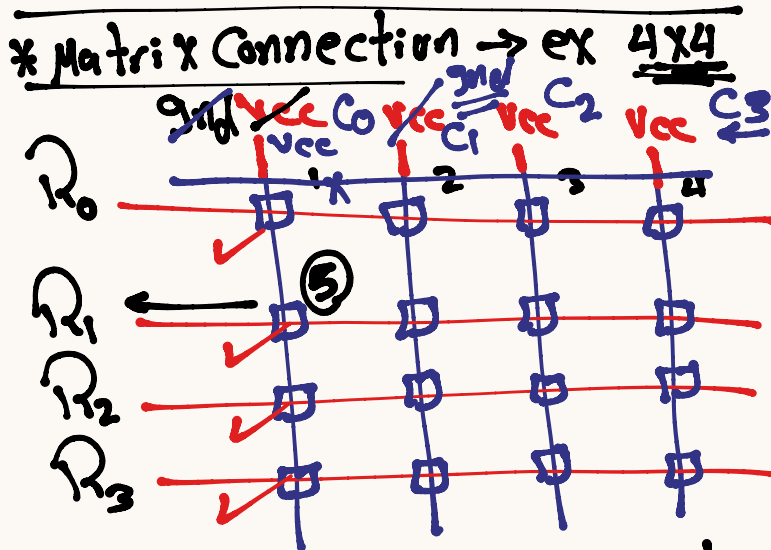


* Key Pad

User → Input Number
 some Button.
 Push Button.

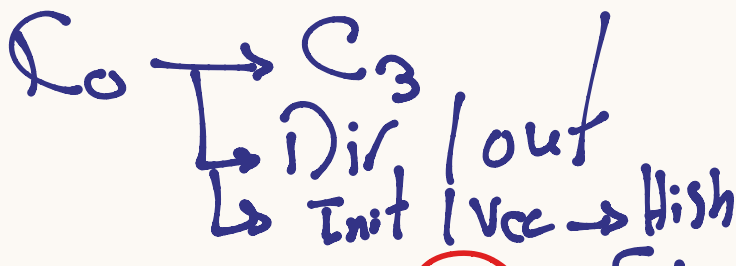
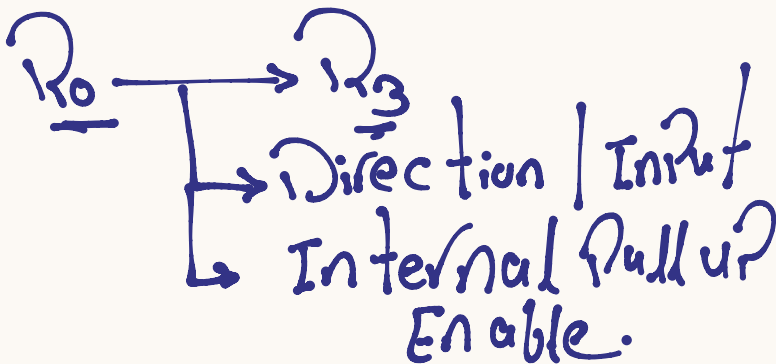
* → 16 Button → 16 pin from mcu to Control.

* → Connect Button by Matrix way



* Button → Pull up Resistor

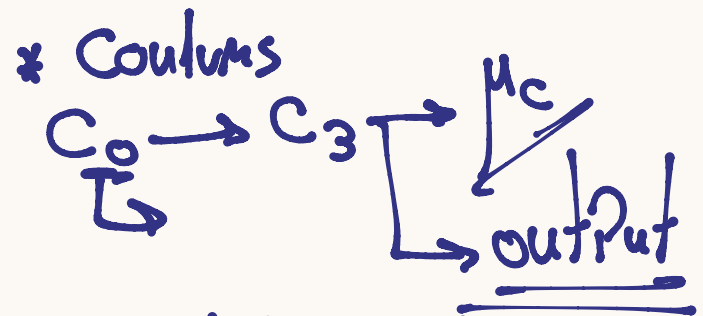
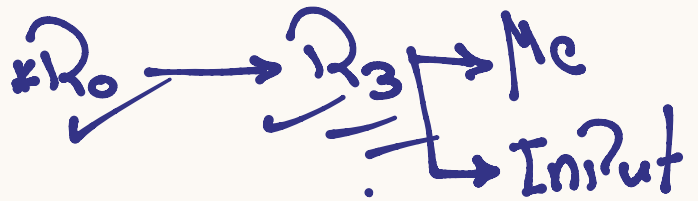
Pressed → 0
 Not Pressed → 1



① → default → Vcc
 ② → fire wire change From Vcc to gnd.

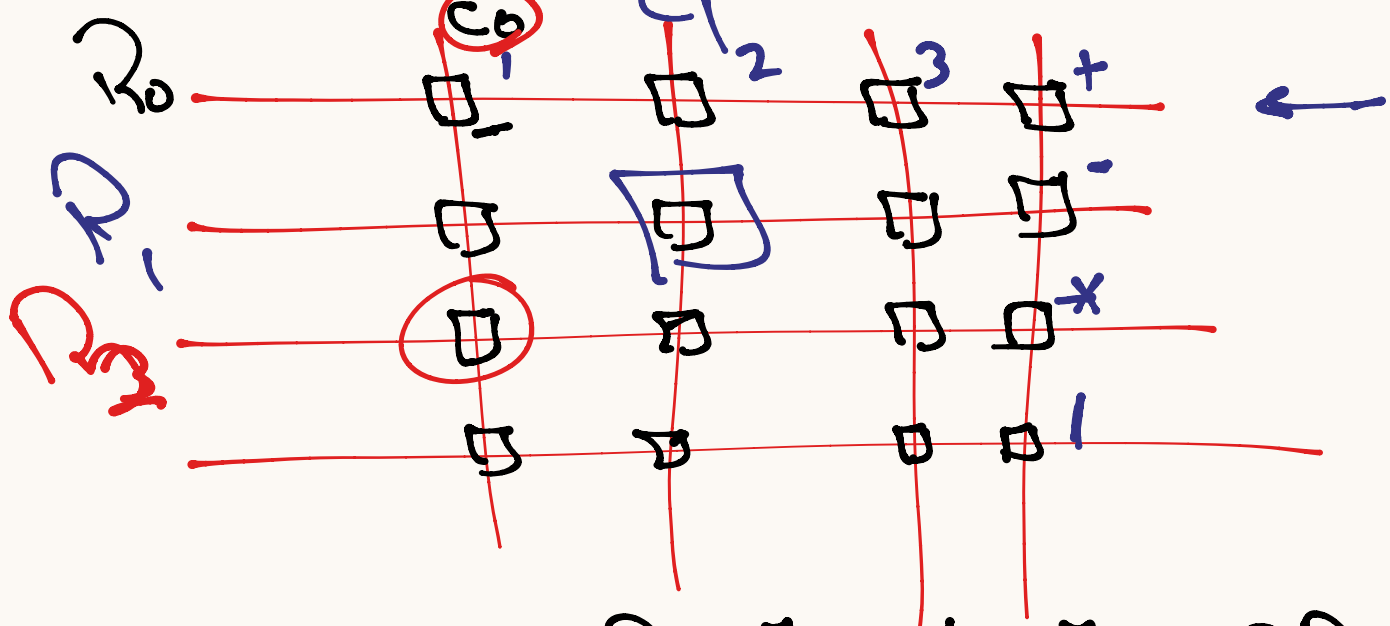
③ → Check on Rows
R0 → R3

if anyone have zero
 if all one return Vcc



void kpd.Init()

R0 → R3
 Input
 Enable Pullup
 C0 → C3
 output
 Initial Vcc(High)



kbd value [Row][Column] → 2D

→ { '1', '2', '3', '+',
 { '4', '5', '6', '-',
 { '7', '8', '9', '*',
 { 'C', '0', '=', '/' }

5,

```
uint8_t kbd_get_kbdvalue ( );
{
  // check on all Button ✓
  // if Not pressed → 0xFF
  // if pressed → kbdvalue
}
```

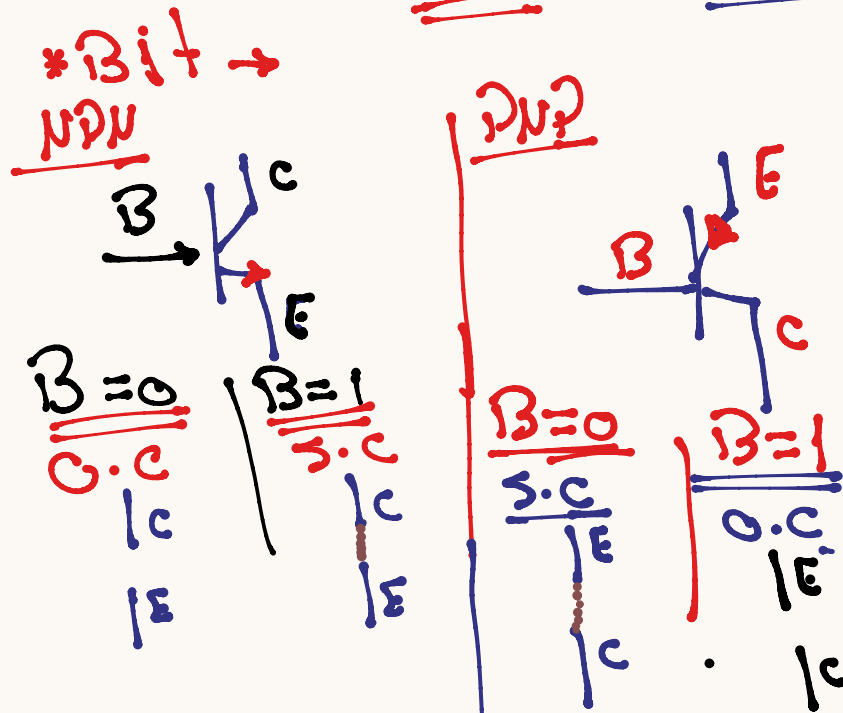
* keyPad → driver

↳ void kpd_Init(void);

↳ uint8_t kpd_GetPressedKey();

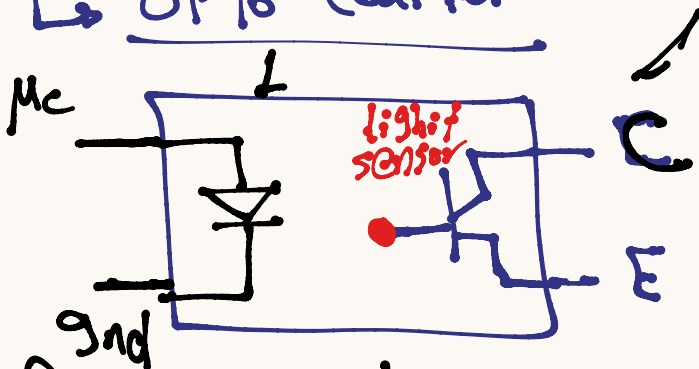
• electrical switch

① transistor
↳ Bjt ↳ Mosfet



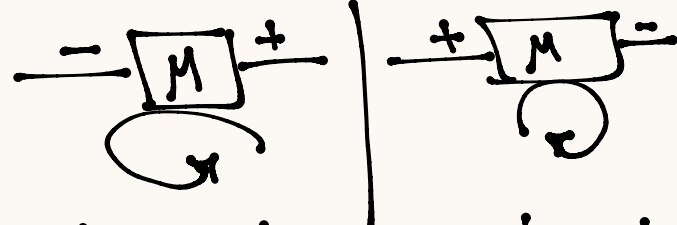
upgrade from transistor

↳ opto coupler

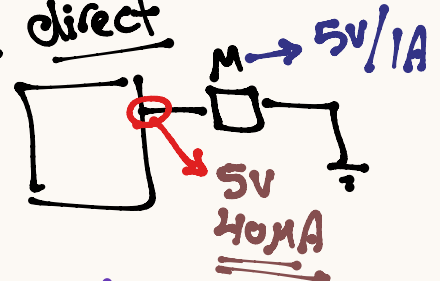


Fully Isolated

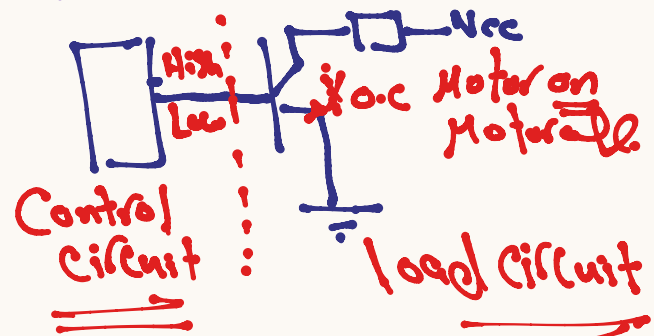
DC Motor ∴ Control



* You Can't Connect Motor with Mc direct



① Connection to Control on Motor as on/off



② Connection to Control on Motor → on/off

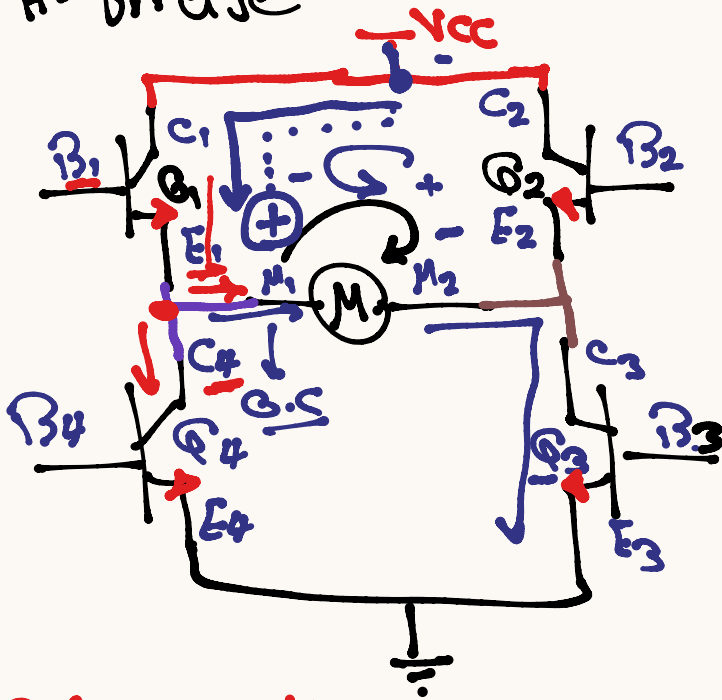
↳ H-bridge circuit

↳ You Can use transistor
↳ You Can use Relay

H-bridge transistor

- ① 4 transistor NPN
- ② 4 transistor PNP
- ③ 2 trans NPN & 2 trans PNP

* H-bridge



- * $C_1 \& C_2 \rightarrow V_{cc}$
- * $E_4 \& E_3 \rightarrow \text{Gnd}$
- * $E_1 \& C_4 \rightarrow M_1$
- * $E_2 \& C_3 \rightarrow M_2$
- * $B_1 \& B_3 \rightarrow A_0$
- * $B_2 \& B_4 \rightarrow A_1$

* O.P.P

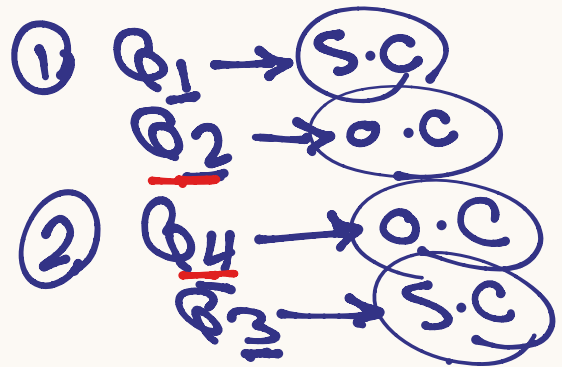
$Q_1 \& Q_2 \& Q_3 \& Q_4$

→ must be O.C

$B_1 \& B_3 \rightarrow A_0 \rightarrow 0$

$B_2 \& B_4 \rightarrow A_1 \rightarrow 0$

CW



$B_1 \& B_3 \rightarrow A_0 \rightarrow \text{High}$
 $B_2 \& B_4 \rightarrow A_1 \rightarrow \text{Low}$

CCW

$Q_1 : \text{O.C}$ $Q_2 : \text{S.C}$
 $Q_3 : \text{O.C}$ $Q_4 : \text{S.C}$

$B_1 \& B_3 \rightarrow A_0 \rightarrow \text{Low}$
 $B_2 \& B_4 \rightarrow A_1 \rightarrow \text{High}$

Note!: - four Code depend on !: transistor type
order

